Mid-tern, Catch-up

20211060 오서현

https://github.com/hyunneee/hyunneee.github.io

in [VMP_midterm catch-up] branch

5. A robot of 5 arms.

```python
def main():
    canvas = np.zeros( (height, width, 3), dtype='uint8')

    deg1 = 0
    deg2 = 0
    deg3 = 0
    v1 = 1
    v2 = 2
    v3 = 3

    a, b = width/2, height*2/3
    w,h = 140, 70

    while True:
        R1 = makeRmat(deg1)
        R2 = makeRmat(deg2)
        R3 = makeRmat(deg3)

    # get ngon points
        rect = getrectangle(h)
        rect2 = getrectangle(h)
        rect3 = getrectangle(h)
        rect4 = getrectangle(h)

    # drawing rect1
        H1 = makeTmat(a,b) @ makeRmat(-90) @ makeTmat(0,-h/2)
        rect = H1 @ rect.T
        rect = rect.T
        rect = rect.astype('int')
        drawPolygon(canvas,rect,(0,0,255),axis = False)

    # drawing rect2
        H2 = makeTmat(w,0) @ makeTmat(0,h/2) @ makeRmat(deg1) @ makeTmat(0,-h/2)
        rect2 = H1 @ H2 @ rect2.T
        rect2 = rect2.T
        rect2 = rect2.astype('int')
        drawPolygon(canvas,rect2,(255,0,255),axis = False)
```

각도

각도조절용

각도만큼 회전시키기

3번 만들

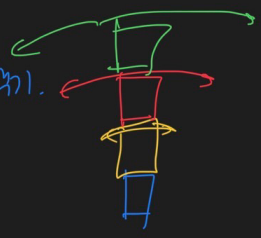
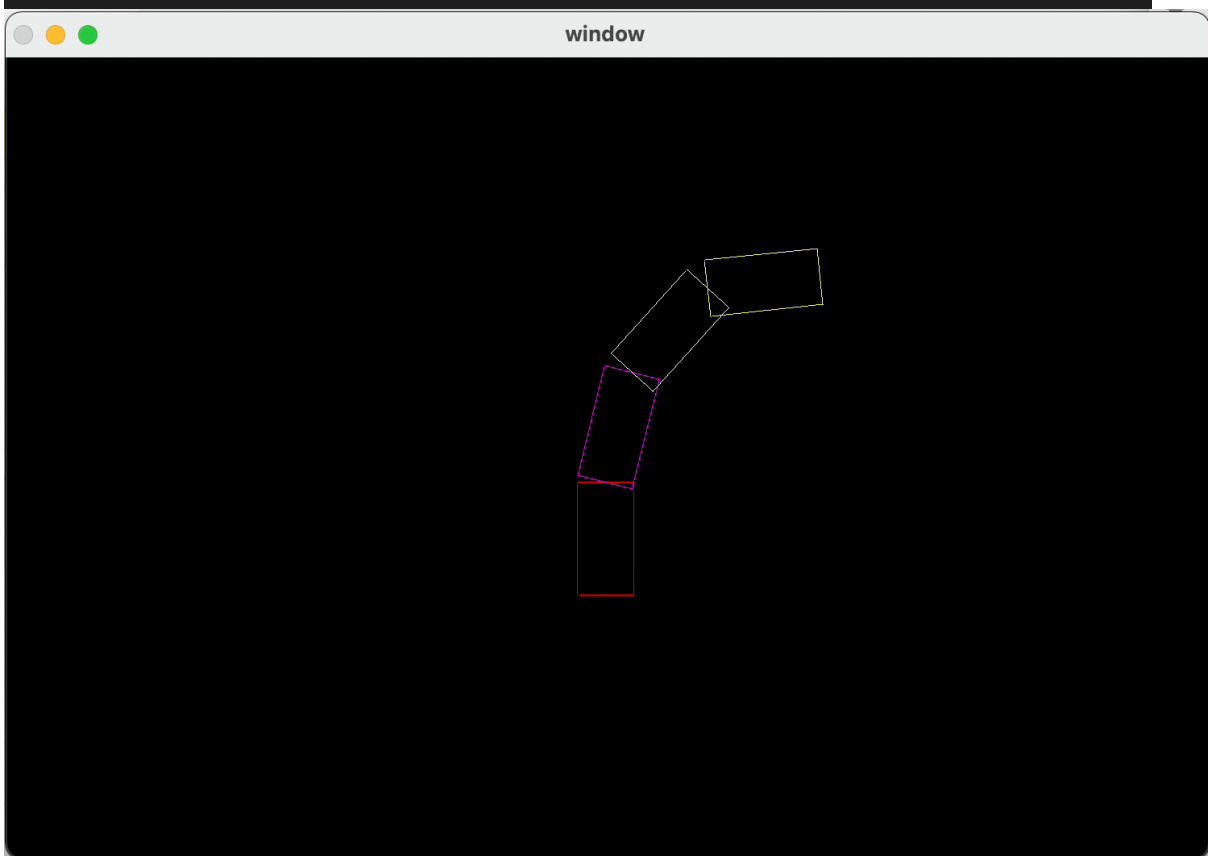
+w

deg1 만큼 회전

```python
    # drawing rect3
        H3 = makeTmat(w,0) @ makeTmat(0,h/2) @ makeRmat(deg2) @ makeTmat(0,-h/2)
        rect3 = H1 @ H2 @ H3 @ rect3.T
        rect3 = rect3.T
        rect3 = rect3.astype('int')
        drawPolygon(canvas,rect3,(255,255,255),axis = False)

    # drawing rect4
        H4 = makeTmat(w,0) @ makeTmat(0,h/2) @ makeRmat(deg3) @ makeTmat(0,-h/2)
        rect4 = H1 @ H2 @ H3 @ H4 @ rect4.T
        rect4 = rect4.T
        rect4 = rect4.astype('int')
        drawPolygon(canvas,rect4,(0,255,255),axis = False)

        if deg1 >= 15 or deg1<= -15:
            v1 = -v1
        if deg2 >= 30 or deg2 <= -30:
            v2 = -v2
        if deg3 >= 45 or deg3 <= -45:
            v3 = -v3

        deg1 += v1
        deg2 += v2
        deg3 += v3
        cv2.imshow("window", canvas)
        canvas[:, :, :] = 0

        if cv2.waitKey(20) == 27: break
#
if __name__ == "__main__": # __
    main()
```

$-15 \le deg1 \le 15$면 부호바꿈.

$-30 \le deg2 \le 30$ 〃

$-45 \le deg3 \le 45$ 〃

지우기